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STORAGE SYSTEM AND PATH MANAGEMENT METHOD

Abstract

A first management device is provided that sets, when a second storage node is increased/decreased at a secondary site, a new remote copy path between a first storage node provided with a primary volume associated with a secondary volume moved in accordance with the increase/decrease, and deletes an original remote copy path set between the second storage that is a movement source of the secondary volume moved in accordance with the increase/decrease and the first storage node.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a storage system and a path management method, and is preferably applied to, for example, a storage system configured to perform data protection by remote copy among a plurality of sites.

2. Description of the Related Art

[0002] Conventionally, a storage system including a plurality of storage nodes is known. For example, predetermined software is executed in each storage node to provide a storage system.

[0003] In addition, there is a remote copy function as a technique for replicating a storage system among a plurality of geographically separated data centers in order to continue business even when a disaster occurs. In a storage system including a plurality of nodes and equipped with a remote copy function, a site that processes a business application at a normal time is referred to as a primary site, and a site that is switched from the primary site and operated when a failure of the entire site occurs on the primary site side and the storage system is stopped is referred to as a secondary site.

[0004] For example, US 2018/0032254 A discloses a technique in which, in a configuration in which a secondary site is configured by a plurality of storage devices, when a remote copy pair is formed between a primary site and a secondary site, a storage device on the secondary site side is selected so as to satisfy performance and capacity requirements of the primary site to establish the pair.

SUMMARY OF THE INVENTION

[0005] Meanwhile, in a storage system using a software defined storage (SDS), scale-out is performed by increasing a storage node on which the SDS is mounted in a case where performance or capacity is insufficient. At this time, in order to improve performance and reliability, software (hereinafter referred to as storage control software) that functions as a storage controller of the SDS arranged in another storage node, the storage control software, and a logical volume owned by the storage control software are rearranged in an increased storage node (hereinafter referred to as an increased storage node). Note that the “ownership” of the logical volume is a right to manage the logical volume and perform I/O process on the logical volume. Only the storage control node having the ownership of the logical volume can perform I/O process on the logical volume.

[0006] Therefore, when the secondary volume of the logical volume of the copy source (hereinafter referred to as a primary volume), and the logical volume of the copy destination (hereinafter referred to as a secondary volume) having the remote copy configuration is moved to the increased storage node due to scale-out at the secondary site, it is necessary to reset the path for remote copy (hereinafter referred to as a remote copy path) between the storage node on the primary site side where the primary volume is created and the increased storage node which is the movement destination of the secondary volume, but US 2018/0032254 A does not disclose the resetting of the remote copy path in such a situation. However, in such a situation, in a case where resetting of the remote copy path is not performed, there is a problem that performance of the remote copy process deteriorates although a specific reason will be described later.

[0007] Furthermore, when scaling in with a storage system using SDS, it is necessary to rearrange the storage control software arranged in the storage node to be decreased and the logical volume owned by the storage control software in another storage node.

[0008] For this reason, in a case where the secondary volume of the primary volume and the secondary volume having the remote copy configuration is arranged in the decreased storage node, it is necessary to reset the remote copy path between the storage node of the movement destination of the secondary volume and the storage node on the primary site side where the primary volume

corresponding to the secondary volume is created, but resetting of the remote copy path in such a situation is not disclosed in US 2018/0032254 A. However, in such a situation as well, in a case where resetting of the remote copy path is not performed, there is a problem that performance of the remote copy process deteriorates.

[0009] The present invention has been made in view of the above points, and aims to propose a storage system and a path management method capable of preventing performance deterioration of the remote copy process after increase/decrease of storage nodes.

[0010] In the present invention for solving the above problem, there is provided a storage system including one or a plurality of first storage nodes installed at a primary site and a plurality of second storage nodes installed at a secondary site, where the first storage node is provided with a primary volume provided to a host device as a storage area for reading and writing data, the second storage node is provided with a secondary volume as a storage area for backing up the data written in the corresponding primary volume, a remote copy path for remotely copying the data written in the primary volume to the secondary volume is set between the first storage node and the second storage node provided with the secondary volume corresponding to the primary volume provided in the first storage node, and a first management device that sets, when the second storage node is increased/decreased at the secondary site, a new remote copy path between the first storage node provided with the primary volume associated with the secondary volume moved in accordance with the increase/decrease, and deletes an original remote copy path set between the second storage that is a movement source of the secondary volume moved in accordance with the increase/decrease and the first storage node.

[0011] Furthermore, in the present invention, there is provided a path management method executed in a storage system including one or a plurality of first storage nodes installed at a primary site and a plurality of second storage nodes installed at a secondary site, in which the first storage node is provided with a primary volume provided to a host device as a storage area for reading and writing data, the second storage node is provided with a secondary volume as a storage area for backing up the data written in the corresponding primary volume, a remote copy path for remotely copying the data written in the primary volume to the secondary volume is set between the first storage node and the second storage node provided with the secondary volume corresponding to the primary volume provided in the first storage node, the storage system includes a first management device, a second management device that is disposed at the primary site and manages each of the first storage nodes installed at the primary site, and a third management device that is disposed at the secondary site and manages each of the second storage nodes installed at the secondary site, the method including a first step in which the first device management collects necessary information from the second and/or third management device when the second storage node is increased/decreased at the secondary site; and a second step in which the first management device instructs the second and third management devices to set a new remote copy path between the first storage node provided with the primary volume associated with the secondary volume moved in accordance with the increase/decrease based on the collected information, and delete the original remote copy path set between the second storage of a movement source of the secondary volume moved in accordance with the increase/decrease and the first storage node.

[0012] According to the storage system and the path management method of the present invention, even after the increase/decrease of the second storage node, it is possible to perform remote copy between the second storage node of a movement destination of the secondary volume moved with the increase/decrease and the first storage node in which the primary volume associated with the secondary volume exists via the newly created remote copy path.

[0013] According to the present invention, a storage system and a path management method capable of preventing performance deterioration of a remote copy process after increase/decrease of storage nodes can be realized.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a block diagram illustrating a schematic overall configuration of a storage system according to an embodiment;

[0015] FIG. 2 is a block diagram illustrating a logical configuration of the storage system;

[0016] FIG. 3 is a block diagram illustrating a flow of data copy from a primary volume to a secondary volume;

[0017] FIG. 4 is a block diagram provided for describing an outline of a pair management function according to the present embodiment;

[0018] FIG. 5 is a block diagram provided for describing an outline of a path management function according to the present embodiment;

[0019] FIG. 6 is a block diagram provided for describing an outline of the path management function according to the present embodiment;

[0020] FIG. 7 is a block diagram illustrating various programs and various tables stored in memories of first and second storage nodes as well as a primary site management device and a secondary site management device;

[0021] FIG. 8 is a table illustrating a configuration example of a system configuration management table;

[0022] FIG. 9 is a table illustrating a configuration example of a storage device management table;

[0023] FIG. 10 is a table illustrating a configuration example of a port management table;

[0024] FIG. 11 is a table illustrating a configuration example of a volume management table;

[0025] FIG. 12 is a table illustrating a configuration example of a volume pair management table;

[0026] FIG. 13 is a table illustrating a configuration example of a path management table;

[0027] FIG. 14 is a flowchart illustrating a processing procedure of secondary site management software side storage node increasing process;

[0028] FIG. 15 is a flowchart illustrating a processing procedure of a pair management software side storage node increasing process;

[0029] FIG. 16 is a flowchart illustrating a processing procedure of secondary site management software side storage node decreasing process;

[0030] FIG. 17A is a flowchart illustrating a processing procedure of a pair management software side storage node decreasing process; and

[0031] FIG. 17B is a flowchart illustrating a processing procedure of a management software side storage node decreasing process.

DESCRIPTION OF THE INVENTION

[0032] Hereinafter, one embodiment of the present invention will be described in detail with reference to the drawings.

[0033] In the following description, an “interface device” may be one or more communications interface devices. The one or more communications interface devices may be one or more homogeneous communications interface devices (for example, one or more network interface cards (NIC)), or may be two or more heterogeneous communications interface devices (for example, an NIC and a host bus adapter (HBA)).

[0034] In the following description, a “memory” is one or more memory devices that are an example of one or more storage devices, and may typically be a main storage device. The at least one memory device in the memory may be a volatile memory device or a non-volatile memory device.

[0035] In the following description, a “storage device” may be one or more permanent storage devices that are an example of one or more storage devices. The permanent storage device may typically be a nonvolatile storage device (for example, an auxiliary storage device), and may

specifically be, for example, a hard disk drive (HDD), a solid state drive (SSD), or a non-volatile memory express (NVMe) drive.

[0036] In addition, in the following description, a “processor” may be one or more processor devices. The at least one processor device may typically be a microprocessor device such as a central processing unit (CPU), but may be another type of processor device such as a graphics processing unit (GPU). The at least one processor device may be single-core or multi-core. The at least one processor device may be a processor core. The at least one processor device may be a processor device in a broad sense, such as a hardware circuit (for example, a field-programmable gate array (FPGA), a complex programmable logic device (CPLD), or an application specific integrated circuit (ASIC)) that performs some or all of the processes.

[0037] In addition, in the following description, information from which an output is obtained with respect to an input will be described with an expression such as “X×X table”, but the information may be data of any structure (for example, it may be structured data or unstructured data), or may be a learning model represented by a neural network, a genetic algorithm, or a random forest that generates an output with respect to an input. Therefore, the “x×x table” can be referred to as “xxx information”. Furthermore, in the following description, the configuration of each table is an example, and one table may be divided into two or more tables, or all or a part of two or more tables may be one table.

[0038] In addition, in the following description, there is a case where process is described with a “program” as a subject, but the subject of the process may be a processor (alternatively, a device such as a controller having the processor) since the program is executed by the processor to perform a defined process appropriately using a storage device and/or an interface device. The program may be installed in a device such as a calculator from a program source. The program source may be, for example, a program distribution server or a calculator-readable (for example, non-transitory) recording medium. In the following description, two or more programs may be implemented as one program, or one program may be implemented as two or more programs.

[0039] In addition, in the following description, in a case where the same type of elements are described without being distinguished, a common portion (portion excluding the branch number) of reference numerals including the branch number is used, and in a case where the same type of elements are described while being distinguished, a reference numeral including the branch number is used. For example, in a case where the nodes are described without being particularly distinguished, the nodes are described as “node xx” (xx is an integer of greater than or equal to 0), and in a case where the individual nodes are described while being distinguished, the nodes may be described as “node xx A” and “node xx B”. In addition, an ID (for example, an identification number) of an element may be used as another description method when describing the same type of element in a distinguished manner. Specifically, for example, the above-described “node xx A” and “node XX B” may be described as “node 1” and “node 2”, respectively.

(1) CONFIGURATION OF STORAGE SYSTEM ACCORDING TO PRESENT EMBODIMENT

[0040] In FIG. 1, reference numeral 1 indicates a storage system according to the present embodiment as a whole. In the present storage system 1, a primary site 2 and a secondary site 3 are connected via a first network 4 including a wide area network (WAN) such as the Internet, and a pair management device 5 installed at a place other than the primary site 2 and the secondary site 3 is connected to the first network 4.

[0041] One or a plurality of host devices 10, one or a plurality of first storage nodes 11, and a primary site management device 12 are installed in the primary site 2, and are connected to each other via a second network 13 such as Ethernet.

[0042] A host device 10 is a general-purpose computer device that functions as a high-rank device for the first storage node 11. However, the host device 10 may be a virtual computer device such as a virtual machine. The host device 10 reads and writes data from and to the first storage node 11 via the second network 13 according to a user operation or a request from a mounted application

program **10A** (FIG. **3**).

[0043] The first storage node **11** is a server device that provides a storage area for reading and writing data with respect to the host device **10**, and includes a control unit **21**, a plurality of storage devices **22**, and one or a plurality of ports **23** connected to one another via an internal bus **20**.

[0044] The control unit **21** includes one or more processors **24** and one or more memories **25**. The processor **24** is a device having a function of controlling the operation of the entire first storage node **11**, and includes a central processing unit (CPU), a micro processing unit (MPU), and the like.

[0045] In addition, the memory **25** includes, for example, a volatile semiconductor memory or a nonvolatile semiconductor memory such as a static random access memory (SRAM) or a dynamic random access memory (DRAM), and is used to temporarily store various programs and necessary data. When the processor **24** executes the program stored in the memory **25**, various processes are executed as the entire first storage node **11** as described later.

[0046] The storage device **22** includes a large-capacity nonvolatile storage device such as an HDD, an SSD, or an NVMe. The storage device **22** provides a physical storage area for reading and writing data from and to the host device **10**. The port **23** is interface hardware for connecting the first storage node **11** to the second network **13**.

[0047] The primary site management device **12** is configured by a general-purpose computer device including information processing resources such as a processor and a memory. The primary site management device **12** manages each of the first storage nodes **11** arranged at the primary site **2** based on the implemented primary site management software (hereinafter referred to as primary site management software) **26**.

[0048] In the secondary site **3**, a plurality of second storage nodes **30** and the secondary site management device **31** are installed, and are connected to each other via a third network **32** such as Ethernet. The second storage node **30** is a server device that provides a storage area for backing up data written to the first storage node **11**. Since the second storage node **30** has a configuration similar to that of the first storage node **11**, a detailed description thereof will be omitted here.

[0049] The second storage node **30** of the secondary site **3** is grouped and managed in a group called a cluster **33** together with other one or a plurality of second storage nodes **30**. In the example of FIG. **1**, a case where only one cluster **33** is set in the secondary site **3** is illustrated, but a plurality of clusters **33** may be set in the secondary site **3**.

[0050] The secondary site management device **31** is configured by a general-purpose computer device including information processing resources such as a processor and a memory. The secondary site management device **31** manages each of the second storage nodes **30** arranged in the secondary site **3** based on the implemented secondary site management software (hereinafter referred to as secondary site management software) **34**.

[0051] The pair management device **5** is also configured by a general-purpose computer device or the like including information processing resources such as a processor and a memory. The pair management device **5** performs a process such as setting and deletion of the remote copy path **7** (FIG. **2**) between the first storage node **11** in the primary site **2** and the second storage node **30** in the secondary site **3** based on the implemented pair management software (hereinafter referred to as pair management software) **6**.

[0052] FIG. **2** is a block diagram illustrating a logical configuration of the storage system **1**. As illustrated in FIG. **2**, one or a plurality of primary volumes PVOL to be provided to the host device **10** (FIG. **1**) as a storage area for reading and writing data is created in each first storage node **11** in the primary site **2**. In addition, in each of the first storage nodes **11**, one or a plurality of storage control software **35** to be described later that manages these primary volumes PVOL is also arranged.

[0053] On the other hand, one or a plurality of storage control software **35** is arranged in each of the second storage nodes **30** of the secondary site **3**. The storage control software **35** is software that functions as a storage controller of the SDS as described above. Each storage control software

35 is managed as one group (hereinafter referred to as a redundancy group) for redundancy together with the storage control software 35 arranged in other one or a plurality of second storage nodes 30 different from each other.

[0054] In the redundancy group, one storage control software 35 is set to a state (a state of a currently used system, hereinafter referred to as an active mode) in which an input/output (I/O) request from the first storage node 11 in the primary site 2 can be accepted, and the remaining storage control software 35 is set to a state (a state of a standby system, hereinafter referred to as a standby mode) in which an I/O request from the first storage node 11 cannot be accepted.

[0055] For example, FIG. 2 illustrates an example in which a redundancy group is configured by the storage control software 35 called "SCS #M" arranged in the second storage node 30 called "Node 1" and the storage control software 35 called "SCS #M" arranged in the second storage node 11 called "Node M", where one of the storage control software 35 arranged in the second storage node 11 called "Node M" is set to the active mode ("Active"), and the other arranged in the second storage node 11 called "Node 1" is set to the standby mode ("Standby").

[0056] In the other drawings, each storage control software 35 written as "SCS #X" arranged in the different second storage node 30 is the storage control software 35 configuring the same redundancy group "X", and among the storage control software, the storage control software 35 written as "Active" represents the storage control software 35 in the active mode, and the storage control software 35 written as "Standby" represents the storage control software 35 in the standby mode.

[0057] In the redundancy group, when a failure occurs in the storage control software 35 in the active mode or the second storage node 30 on which the storage control software 35 operates, the state of the storage control software 35 that has been set to the standby mode is switched to the active mode. As a result, when the storage control software 35 set to the active mode cannot be operated, the I/O processing executed by the storage control software 35 can be taken over by the storage control software 35 that has been set to the standby mode until then (fail-over function).

[0058] In order to realize such a fail-over function, the storage control software 35 belonging to the same redundancy group always holds configuration information of the same content. The configuration information is information necessary for the storage control software 35 to execute processes related to various functions such as a remote copy function for copying data to a remote location in a synchronous or asynchronous manner for disaster countermeasures, such as mapping information for associating a storage area of the secondary volume SVOL with a storage area of the storage device 22 (FIG. 1), a capacity virtualization function for virtualizing a storage capacity of each storage device 22 (FIG. 1) mounted on the second storage node 30, a hierarchical storage control function for moving data having a high access frequency to a storage area having a higher response speed, a deduplication function for deleting duplicated data from stored data, a data compression function for compressing and storing data, and a snapshot function for holding a state of data at a certain time point.

[0059] Then, when the configuration information of the storage control software 35 in the active mode configuring the redundancy group is updated, a difference between the configuration information of before and after the update is transferred to the other storage control software 35 configuring the redundancy group as differential data, and the configuration information held by the storage control software 35 is updated by the other storage control software 35 based on the differential data. As a result, the configuration information held by each storage control software 35 constituting the redundancy group is always maintained in a synchronized state.

[0060] As described above, all the storage control software 35 constituting the redundancy group always hold the configuration information having the same contents, and thus, even in a case where a failure occurs in the storage control software 35 set to the active mode or the second storage node 30 in which the storage control software 35 is arranged, or the second storage node 30 is removed, the process executed by the storage control software 35 until then can be immediately taken over

by another storage control software 35 constituting the same redundancy group as the storage control software 35.

[0061] On the other hand, in the secondary site 3, a secondary volume SVOL is created in the second storage node 30 in association with each primary volume PVOL created in the primary site 2. The secondary volume SVOL is a logical volume for backing up data stored in the corresponding primary volume PVOL.

[0062] Each secondary volume SVOL corresponding to one primary volume PVOL is created in the second storage node 30 in which the storage control software 35 is arranged in association with each storage control software 35 constituting the same redundancy group.

[0063] Therefore, in a case where the redundancy group is configured by two pieces of storage control software 35 as illustrated in FIG. 2, two secondary volumes SVOL are created in different second storage nodes 30 in the secondary site 2 with respect to one primary volume PVOL, and these secondary volumes SVOL are respectively associated with the storage control software 35 configuring the same redundancy group.

[0064] In addition, a remote copy path 7 is set between the first storage node 11 of the primary site 2 in which the primary volume PVOL is provided and each second storage node 30 in which the secondary volume SVOL is created in correspondence with the primary volume PVOL.

[0065] Then, the data written in the primary volume PVOL is transferred to the second storage node 30 via the remote copy path 7, among the remote copy paths 7, connecting the first storage node 11 in which the primary volume PVOL is created and the second storage node 30 in which the secondary volume SVOL associated with the storage control software 35 in the active mode among the corresponding secondary volumes SVOL is created.

[0066] Furthermore, the data transferred to the second storage node 30 is then stored in the secondary volume SVOL created in association with the primary volume PVOL in the second storage node 30 under the control of the storage control software 35 in the active mode.

[0067] Furthermore, the data is transferred from the second storage node 30 in which the storage control software 35 in the active mode is arranged to each second storage node 30 in which each storage control software 35 in the standby mode constituting the same redundancy group as the storage control software 35 is arranged via the third network 32 (FIG. 1). Then, the data is also stored in the secondary volume SVOL created in correspondence with the primary volume PVOL under the control of the storage control software 35 in the second storage node 30.

[0068] In the following description, the storage control software 35 in the active mode is appropriately referred to as active storage control software 35, and the storage control software 35 in the standby mode is appropriately referred to as standby storage control software 35. In addition, the secondary volume SVOL managed by the active storage control software 35 is appropriately referred to as an active secondary volume SVOL, and the secondary volume SVOL managed by the standby storage control software 35 is appropriately referred to as a standby secondary volume SVOL.

[0069] Furthermore, in the following description, a combination of the primary volume PVOL and the secondary volume SVOL created in correspondence with the primary volume PVOL is referred to as a volume pair. In a case where a plurality of secondary volumes SVOL are created by redundancy with respect to the primary volume PVOL, each of the combinations of the primary volume PVOL and these secondary volumes SVOL is a volume pair.

[0070] FIG. 3 illustrates a specific flow of remote copy of data from the primary volume PVOL to each corresponding secondary volume SVOL as described above. In the case of the storage system 1 of the present embodiment, the remote copy between the primary volume PVOL and the secondary volume SVOL is performed asynchronously with the timing at which the host device 10 writes data to the primary volume PVOL.

[0071] In practice, in the case of the present storage system 1, in each of the first storage nodes 11 in which the primary volume PVOL to be backed up in the primary site 2 is created, a logical

volume called a journal volume PJVOL is created in correspondence with the primary volume PVOL. Hereinafter, the journal volume PJVOL is referred to as a primary journal volume PJVOL. One primary journal volume PJVOL may be created for one primary volume PVOL, or may be created for a plurality of primary volumes PVOL.

[0072] In addition, journal volumes SJVOL are also created in association with the respective secondary volumes SVOL in the respective second storage nodes **30** of the secondary site **3** in which the respective secondary volumes SVOL created in association with the primary volume PVOL are created. Hereinafter, the journal volume SJVOL is referred to as a secondary journal volume SJVOL. One secondary journal volume SJVOL may be created for one secondary volume SVOL, or may be created for a plurality of secondary volumes SVOL.

[0073] The secondary journal volume SJVOL and the secondary volume SVOL are created by cutting out from a storage area called a pool PL in which storage areas provided by the respective storage devices **22** mounted on the second storage node are gathered. The same applies to the primary journal volume PJVOL and the primary volume PVOL.

[0074] Then, when the host device **10** writes data (“A” and “B” in FIG. **3**) to the primary volume PVOL, metadata such as the position where the data is written in the primary volume PVOL, the data length and the sequence number of the data, and the like are added to the data to generate journal data JNL, and the generated journal data JNL is written to the primary journal volume PJVOL associated with the primary volume PVOL.

[0075] Furthermore, the journal data JNL written to the primary journal volume PJVOL is transmitted to the second storage node **30** in which the active secondary volume SVOL corresponding to the primary volume PVOL is created at a predetermined timing such as, for example, when the load of the first storage node **11** is low, and thereafter, is stored in the secondary journal volume SJVOL created in the second storage node **30** in association with the active secondary volume SVOL.

[0076] Then, the journal data JNL stored in the secondary journal volume SJVOL is read from the secondary journal volume SJVOL by the active storage control software **35** having the ownership of the active secondary volume SVOL. Then, the active storage control software **35** removes the metadata from the read journal data JNL to extract only the data written in the corresponding primary volume PVOL and store the extracted data in the active secondary volume SVOL.

[0077] Specifically, the active storage control software **35** acquires the position where the data is written in the corresponding primary volume PVOL included in the metadata removed from the journal data JNL. In addition, the active storage control software **35** specifies the storage device **22** providing the storage area at the position on the active secondary volume SVOL constituting the volume pair with the primary volume PVOL, and writes the data to the storage device **22**.

[0078] Furthermore, the active storage control software **35** transfers the journal data JNL stored in the secondary journal volume SJVOL to the standby storage control software **35** associated with the standby secondary volume SVOL in each second storage node **30** in which the standby secondary volume SVOL created in association with the primary volume PVOL exists. Then, the standby storage control software **35** that has received the journal data stores the received journal data in the secondary journal volume SJVOL associated with the standby secondary volume SVOL.

[0079] In addition, the active storage control software **35** transfers the data stored in the active secondary volume SVOL as described above (the data written in the corresponding primary volume PVOL) to the standby storage control software **35** associated with the standby secondary volume SVOL in each of the second storage nodes **30** in which the corresponding standby secondary volume SVOL exists. At this time, the active storage control software **35** associated with the active secondary volume SVOL also notifies the position where the data is written in the active secondary volume SVOL.

[0080] Thus, the standby storage control software **35** that has received the data stores the received data in the standby secondary volume SVOL. Specifically, the standby storage control software **35**

specifies a storage device **22** providing a storage area at a position (a position where the data is written in the corresponding primary volume PVOL) notified from the active storage control software **35** associated with the active secondary volume SVOL on the standby secondary volume SVOL, and writes the data in the storage device **22**. As a result, such data is made redundant on the secondary site **3** side.

(2) PATH MANAGEMENT FUNCTION ACCORDING TO THE PRESENT EMBODIMENT

(2-1) Overview of Path Management Function According to the Present Embodiment

[0081] Next, a path management function in the storage system **1** will be described. The path management function is a function of, when the second storage node **30** is increased/decreased in the secondary site **3**, resetting a new remote copy path **7** to the second storage node **30** of a rearrangement destination between the storage control software **35** rearranged accordingly and the secondary volume SVOL of which the storage control software **35** has the ownership, and deleting the corresponding original remote copy path **7** that has been set to the second storage node **30** of a rearrangement source.

[0082] Hereinafter, transferring the storage control software **35** and the secondary volume SVOL from a certain second storage node **30** to another second storage node **30** is referred to as “rearrangement” or “movement”. Hereinafter, “rearrangement” and “movement” are used interchangeably.

[0083] For example, when the second storage node **30** of “M+1” is increased as illustrated in FIG. **4** with respect to the cluster **33** (FIG. **1**) of the secondary site **3** having the configuration illustrated in FIG. **2**, the active storage control software **35** (“SCS #M+1 Active” in FIG. **4**) is arranged in the increased second storage node (increased storage node) **30**.

[0084] By the way, when the number of storage control software **35** configuring the redundancy group in the cluster **33** is “2” (when the redundancy degree in the cluster **33** is “2”), it is necessary to arrange the active storage control software **35** arranged in the increased storage node **30** and one standby storage control software **35** configuring the redundancy group in any second storage node **30** other than the increased storage node **30** in the cluster **33**.

[0085] In this case, in consideration of load balancing of each of the second storage nodes **30** in the secondary site **3**, the number of storage control software **35** arranged in each of the second storage nodes **30** is preferably the same among the second storage nodes **30**.

[0086] Therefore, in the present embodiment, when arranging the active storage control software **35** arranged in the increased storage node **30** and the one or more standby storage control software **35** configuring the redundancy group in the other second storage node **30**, as illustrated in FIG. **5**, the existing standby storage control software **35** (“SCS #M Standby” in FIG. **5**) arranged in any of the second storage nodes **30**, the standby secondary volume SVOL for which the standby storage control software **35** has the ownership, and the configuration information thereof are rearranged in the increased storage node **30**, and then the active storage control software **35** arranged in the increased storage node **30** and the standby storage control software **35** (“SCS #M+1 Standby” in FIG. **5**) configuring the redundancy group are arranged in the second storage node **30**.

[0087] In such a state, when a failure occurs in the active storage control software **35** configuring the same redundancy group as the standby storage control software **35** (hereinafter referred to as rearrangement storage standby control software **35**) rearranged in the increased storage node **30** or the second storage node **30** in which the active storage control software **35** is arranged, the corresponding first storage node **11** in the primary site **2** thereafter starts to transmit the journal data JNL to the second storage node **30** (“Node 1” in FIG. **5**) of the rearrangement source in which the secondary volume SVOL, for which the rearrangement standby storage control software **35** has the ownership, was arranged before the rearrangement.

[0088] However, since the rearrangement standby storage control software **35** and the secondary volume SVOL owned by the rearrangement standby storage control software **35** do not already exist in the second storage node **30** of the rearrangement source, the journal data JNL is transferred

to the increased storage node **30** serving as the rearrangement destination of the rearrangement standby storage control software **35** via the third network **32** (FIG. **1**).

[0089] However, when such transfer occurs, there is a problem that the response on the secondary site **3** side with respect to the remote copy between the primary site **2** and the secondary site **3** is delayed, and furthermore, an extra load for the transfer of the journal data JNL is generated in the second storage node **30** of the rearrangement source of the rearrangement standby storage control software **35**.

[0090] On the other hand, for example, as illustrated in FIG. **6**, when the second storage node **30** of “M” is decreased from the cluster **33** of the secondary site **3** having the configuration illustrated in FIG. **2**, it is necessary to rearrange the storage control software **35** (“SCS #M Active” and “SCS #M-1 Standby” in FIG. **6**) existing in the second storage node (decreased storage node) **30** to another second storage node **30** and move the secondary volume SVOL managed by the storage control software **35** to the second storage node **30**.

[0091] Therefore, in the present embodiment, the active secondary volume SVOL for which the active storage control software **35** arranged in the decreased storage node **30** has the ownership is moved to the other second storage node **30**, and the ownership of the active secondary volume SVOL is transferred from the active storage control software **35** of the decreased storage node **30** to the active storage control software **35** in the second storage node **30** to which the active secondary volume SVOL has been moved. Furthermore, for the standby storage control software **35** arranged in the decreased storage node **30**, the standby storage control software **35**, the standby secondary volume SVOL for which the standby storage control software **35** has the ownership, and the configuration information thereof are rearranged in any other second storage node **30**.

[0092] In such a state, in a case where the decreased storage node **30** is actually decreased, the first storage node **11** in which the primary volume PVOL corresponding to the active secondary volume SVOL owned by the active storage control software **35** in the decreased storage node **30** is arranged recognizes that a failure has occurred in the decreased storage node **30**. Then, the journal data JNL of the data written in the primary volume PVOL which has been transmitted to the decreased storage node **30** so far is transmitted to the second storage node **30** (“Node 1” in FIG. **6**) in which the standby storage control software **35** (“SCS #M Standby” in FIG. **6**) constituting the same redundancy group as the active storage control software **35** is arranged.

[0093] Therefore, the second storage node **30** that has received the journal data JNL transfers the journal data JNL to the second storage node **30** (“NodeM -1” in FIG. **6**) of the rearrangement destination of the active storage control software **35**.

[0094] However, when such transfer occurs, there is a problem that the response on the secondary site **3** side to the remote copy between the primary site **2** and the secondary site **3** is delayed, and furthermore, an extra load for the transfer occurs in the second storage node **30** that transfers the journal data JNL.

[0095] Therefore, in the storage system **1** of the present embodiment, when the second storage node **30** is increased/decreased in the secondary site **3**, a new remote copy path **7** is reset to the second storage node **30** of a rearrangement destination between the storage control software **35** rearranged accordingly and the secondary volume SVOL for which the storage control software **35** has the ownership, and the original remote copy path **7** set to the second storage node **30** of the rearrangement source is deleted, thereby suppressing the transfer of the journal data JNL in the secondary site **3** as described above.

[0096] In practice, in the storage system **1**, when the increased storage node **30** is newly installed on the secondary site **3** side, the secondary site management device **31** gives an instruction to the second storage node **30** of a rearrangement source of the standby storage control software **35** and the increased storage node **30** so as to rearrange (copy) the standby storage control software **35** arranged in any of the second storage nodes **30**, the standby secondary volume SVOL for which the standby storage control software **35** has the ownership, and the configuration information thereof to

the increased storage node **30**.

[0097] Note that there is a case where copy or creation of the corresponding secondary journal volume SJVOL is required in the second storage node **30** of the copy destination along with the copy of the secondary volume SVOL between the second storage nodes **30**, but for ease of description and understanding, description of copy or creation of the secondary journal volume SJVOL will all be omitted below.

[0098] Then, when the rearrangement of the standby storage control software **35** and the standby secondary volume SVOL is completed, the secondary site management device **31** transmits, to the pair management device **5**, rearrangement information including information such as the storage control software ID of the standby storage control software **35** of each piece of the rearranged standby storage control software **35**, the volume ID of standby secondary volume SVOL for which each piece of the standby storage control software **35** has the ownership, the node ID of the second storage node **30** of the rearrangement source of each piece of the standby storage control software **35**, and the node ID of the second storage node (increased storage node) **30** of the rearrangement destination of the standby storage control software **35**.

[0099] The pair management device **5** that has received the rearrangement information acquires, for each rearranged standby storage control software **35**, the IP address of each port **23** included in the first storage node **11** in which the primary volume PVOL corresponding to the standby secondary volume SVOL for which the standby storage control software **35** has the ownership and the storage control software **35** having the ownership of the primary volume PVOL are arranged, and the IP address of each port **23** included in the increased storage node **30** from the primary site management device **12**.

[0100] Then, based on the acquired information, the pair management device **5** resets the remote copy path **7** for performing remote copy between the standby secondary volume SVOL for which the rearranged standby storage control software **35** has ownership and the primary volume PVOL associated with the standby secondary volume SVOL between the first storage node **11** and the increased storage node **35** in which the standby storage control software **35** is rearranged.

[0101] In addition, the pair management device **5** deletes the remote copy path **7** for performing remote copy between the primary volume PVOL and the standby secondary volume SVOL set between the first storage node **11** and the second storage node **30** of the rearrangement source of each standby storage control software **35** rearranged in the increased storage node **30**.

[0102] On the other hand, when the information indicating to decrease any of the second storage nodes **30** in the secondary site **3** is given from the user, the secondary site management device **31** gives an instruction to the second storage node **30** and the decreased storage node **30** so that the active secondary volume SVOL for which the active storage control software **35** in the second storage node (decreased storage node) **30** has the ownership and its configuration information are moved to the second storage node **30** selected by the secondary site management device **31** other than the decreased storage node **30** and the ownership of the active secondary volume SVOL is transferred to the active storage control software **35** in the second storage node **30**.

[0103] In addition, the secondary site management device **31** gives an instruction to the second storage node **30** and the decreased storage node **30** to rearrange the standby storage control software **35** arranged in the decreased storage node **30**, the standby secondary volume SVOL owned by the standby control software **35**, and the configuration information thereof in one or a plurality of second storage nodes **30** selected by the secondary site management device **31** other than the decreased storage node **30**.

[0104] Then, when the rearrangement of each secondary volume SVOL arranged in the decreased storage node **30** and each storage control software **35** is completed, the secondary site management device **31** transmits, to the pair management device **5**, rearrangement information including information such as the storage control software ID of each rearranged storage control software **35** (also include the active storage control software **35** to which the ownership of the active secondary

volume SVOL is transferred), the volume ID of each secondary volume SVOL for which each storage control software **35** has or had the ownership, the node ID of the second storage node (decreased storage node) **30** of the rearrangement source of each storage control software **35**, the node ID of the second storage node **30** of the rearrangement destination of each storage control software **35** (include transfer of the ownership of the active secondary volume SVOL).

[0105] The pair management device **5** that has received the rearrangement information acquires, for each rearranged storage control software **35**, the IP address of each port **23** included in the first storage node **11** in which the primary volume PVOL corresponding to the secondary volume SVOL for which the storage control software **35** has the ownership and the storage control software **35** having the ownership of the primary volume PVOL are arranged, and the IP address of each port **23** included in the second storage node **30** of the rearrangement destination of the rearranged storage control software **35** from the primary site management device **12**.

[0106] Then, based on the acquired information, the pair management device **5** resets the remote copy path **7** for performing remote copy between the active secondary volume SVOL moved from the decreased storage node **30** and the primary volume PVOL corresponding to the active secondary volume SVOL between each of the second storage nodes **30** of the movement destination of the active secondary volume SVOL and the first storage node **11** in which the primary volume PVOL is arranged.

[0107] In addition, the pair management device **5** resets the remote copy path **7** for performing remote copy between each standby secondary volume SVOL moved from the decreased storage node **30** and each primary volume PVOL corresponding to each standby secondary volume SVOL between the second storage node **30** of the rearrangement destination of each storage control software **35** and the first storage node **11** in which the corresponding primary volume PVOL is arranged.

[0108] Furthermore, the pair management device **5** deletes the remote copy path **7** for performing remote copy between the primary volume PVOL and the active secondary volume SVOL set between the first storage node **11** in which the primary volume PVOL associated with active secondary volume SVOL for which the active storage control software **35** in the decreased storage node **30** has ownership is arranged, and the decreased storage node **30**.

[0109] Similarly, the pair management device **5** deletes the remote copy path **7** for performing remote copy between each primary volume PVOL and each standby secondary volume SVOL set between each first storage node **11** in which each primary volume PVOL associated with each standby secondary volume SVOL in the decreased storage node **30** is arranged and the decreased storage node **30**.

(2-2) Configuration of Table and Program Related to Path Management Function According to the Present Embodiment

[0110] As a means for realizing the path management function according to the present embodiment as described above, as illustrated in FIG. 7, a system configuration management table **40**, a storage device management table **41**, a port management table **42**, a volume management table **43**, a volume pair management table **44**, a path management table **45**, an I/O processing unit **50**, a node increasing processing unit **51**, and a node decreasing processing unit **52** are stored in the memory **25** (FIG. 1) of each of the first and second storage nodes **11** and **30**.

[0111] The primary site management device **12** and the secondary site management device **31** also store a system configuration management table **40**, a storage device management table **41**, a port management table **42**, a volume management table **43**, a volume pair management table **44**, and a path management table **45** similar to those of the I/O processing unit **50**, a node increasing processing unit **51**, and a node decreasing processing unit **52**.

[0112] The system configuration management table **40** is a table for managing all the first and second storage nodes **11** and **30** existing in the storage system **1**, and includes a node ID field **40A**, a state field **40B**, a storage device ID list field **40C**, and a port ID list field **40D** as illustrated in

FIG. 8. In the system configuration management table **40**, one record (row) corresponds to any one of the first or second storage nodes **11** and **30** existing in the storage system **1**.

[0113] Then, in the node ID field **40A**, an identifier (node ID) unique to the first or second storage node **11** or **30** in the storage system **1** assigned to the corresponding first or second storage node **11** or **30** is stored. In the state field **40B**, the current state of the first or second storage node **11** or **30** is stored. As the state of the first or second storage node **11** or **30**, there are “Failure” meaning that a failure has occurred, “Normal” meaning that it is a normal state in which no failure has occurred, and the like.

[0114] In the storage device ID list field **40C**, all the identifiers (storage device IDs) unique to the respective **22** storage devices **22** (FIG. 1) in the storage system **1** assigned to each storage device **22** mounted on the corresponding first or second storage nodes **11** and **30** are stored. In the port ID list field **40D**, all the identifiers (port IDs) unique to the ports **23** (FIG. 1) of the corresponding first or second storage nodes **11** and **30** in the storage system **1** assigned to each of the ports **23** are stored.

[0115] Therefore, in the case of the example of FIG. 8, for example, the current state of the first or second storage nodes **11** and **30** to which the node ID of “0” is assigned is normal (“Normal”), and the first or second storage nodes **11** and **30** are mounted with the storage devices **22** to which the storage device IDs of “0”, “1”, and “2” are assigned, respectively. FIG. 8 also illustrates that the first or second storage nodes **11** and **30** include a port **23** assigned with a port ID of “0” and a port **23** assigned with a port ID of “1”.

[0116] The storage device management table **41** is a table used to manage the storage devices **22** mounted on the first and second storage nodes **11** existing in the storage system **1**, and includes a storage device ID field **41A**, a state field **41B**, and a size field **41C** as illustrated in FIG. 9. In the storage device management table **41**, one record corresponds to one storage device **22** mounted on any one of the first or second storage nodes **11** and **30**.

[0117] Then, the storage device ID of the corresponding storage device **22** is stored in the storage device ID field **41A**, and the current state of the storage device **22** is stored in the state field **41B**. The state of the storage device **22** includes “Failure” meaning that a failure has occurred, “Normal” meaning that it is a normal state in which no failure has occurred, and the like. In the size field **41C**, the capacity of the storage device **22** is stored.

[0118] Therefore, in the case of the example of FIG. 9, for example, it is indicated that the current state of the storage device **22** to which the storage device ID of “0” is assigned is normal (“Normal”) and the capacity thereof is “1400 GB”.

[0119] The port management table **42** is a table used to manage the ports **23** included in each of the first and second storage nodes **11** existing in the storage system **1**, and includes a port ID field **42A**, a state field **42B**, and an IP address field **42C** as illustrated in FIG. 10. In the port management table **42**, one record corresponds to one port **23** included in any of the first or second storage nodes **11** and **30**.

[0120] Then, the port ID of the corresponding port **23** is stored in the port ID field **42A**, and the current state of the port **23** is stored in the state field **42B**. The state of the port **23** includes “Failure” meaning that a failure has occurred, “Normal” meaning that it is a normal state in which no failure has occurred, and the like. The IP address of the port **23** is stored in the IP address field **42C**.

[0121] Therefore, in the case of the example in FIG. 10, for example, it is indicated that the current state of the port **23** to which the port ID of “0” is assigned is normal (“Normal”), and the IP address of the port **23** is “172.12.16.200”.

[0122] The volume management table **43** is a table used to manage each logical volume created in the storage system **1**, and includes a volume ID field **43A**, an owner ID field **43B**, a backward destination ID field **43C**, a size field **43D**, an attribute field **43E**, and a node ID field **43F** as illustrated in FIG. 11. In the volume management table **43**, one record corresponds to one logical

volume created in the storage system **1**.

[0123] An identifier (volume ID) unique to the corresponding logical volume in the storage system **1** and assigned to the corresponding logical volume is stored in the volume ID field **43A**, and a node ID of the first or second storage node **11** or **30** in which the logical volume is created is stored in the node ID field **43F**.

[0124] In the owner ID field **43B**, an identifier (storage control software ID) of the storage control software **35** having the ownership of the logical volume is stored, and a storage control software ID of the storage control software **35** of the backward destination configuring the same redundancy group together with the storage control software **35** is stored.

[0125] Furthermore, the capacity of the corresponding logical volume is stored in the size field **43D**, and the attribute of the logical volume is stored in the attribute field **43E**. The attributes of the logical volume include “normal VOL” meaning that remote copy is not set for any logical volume of the logical volume, “PVOL” meaning that the logical volume is the primary volume PVOL of the volume pair, “SVOL” meaning that the logical volume is the secondary volume SVOL of the volume pair, and “JNLVOL” meaning that the logical volume is the journal volume.

[0126] Therefore, in the case of the example of FIG. **11**, for example, it is indicated that a logical volume to which a volume ID of “0” is assigned is a logical volume created in the first or second storage nodes **11** and **30** to which a node ID of “0” is assigned, the capacity thereof is “500 GB”, and the attribute thereof is “normal VOL”.

[0127] In FIG. **11**, the logical volume indicates that the storage control software **35** having a storage control software ID of “0” has the ownership, and the storage control software **3** configures the same redundancy group as the storage control software **35** having a storage control software ID of “0”.

[0128] The volume pair management table **44** is a table used to manage each volume pair defined in the storage system **1**, and includes a volume pair ID field **44A**, a primary volume ID field **44B**, a primary journal volume ID field **44C**, a secondary journal volume ID field **44D**, a secondary volume ID field **44E**, a path ID field **44F**, a state field **44G**, and an attribute field **44H** as illustrated in FIG. **12**. In the volume pair management table **44**, one record corresponds to one volume pair defined in the storage system **1**.

[0129] In the volume pair ID field **44A**, an identifier (volume pair ID) unique to the corresponding volume pair assigned to the corresponding volume pair in the storage system **1** is stored.

[0130] In addition, the volume ID of the primary volume PVOL in the volume pair is stored in the primary volume ID field **44B**, and the volume ID of the primary journal volume PJVOL (journal data JNL (FIG. **3**) of data stored in the primary volume PVOL is stored) associated with the primary volume PVOL is stored in the primary journal volume ID field **44C**.

[0131] Furthermore, the secondary journal volume field **44D** stores the volume ID of the secondary journal volume SJVOL to be the transfer destination of the journal data JNL stored in the primary journal volume PJVOL, and the secondary volume ID field **44E** stores the volume ID of the secondary volume SVOL in the corresponding volume pair.

[0132] Furthermore, the path ID field **44F**, an identifier (path ID) unique to the remote copy path **7** assigned to the remote copy path **7** used for remote copy of data from the primary volume PVOL in which the volume ID is stored in the primary volume ID field **44B** to the secondary volume SVOL in which the volume ID is stored in the secondary volume ID field **44E** is stored.

[0133] Furthermore, the current state of the corresponding volume pair is stored in the state field **44G**. Examples of the state of the volume pair include “Copying” indicating a state in which remote copy is in progress, “Suspend” indicating a state in which remote copy is paused, and “Normal” indicating a normal state but a state in which remote copy is not being performed.

[0134] Furthermore, the attribute field **44H** stores the attribute of the secondary volume SVOL in the copy pair. The attribute includes “Active” indicating that the corresponding secondary volume SVOL is the active secondary volume SVOL and “Standby” indicating that the secondary volume

SVOL is the standby secondary volume SVOL. The first storage node **11** refers to the volume pair management table **44**, and transmits the journal data JNL to the corresponding second storage node **30** using the remote copy path **7** connected to the secondary volume SVOL whose attribute of the secondary volume SVOL is “Active”.

[0135] Therefore, in the case of the example of FIG. **12**, for example, the volume pair to which the volume pair ID of “O” is assigned is a volume pair in which the logical volume to which the volume ID of “1” is assigned is the primary volume PVOL and the logical volume to which the volume ID of “101” is assigned is the secondary volume SVOL.

[0136] In addition, FIG. **12** also illustrates that the data stored in the primary volume PVOL is transmitted from the primary volume PVOL to the secondary volume SVOL through the primary journal volume PJVOL to which the volume ID of “2” is assigned in the primary site **2** and the secondary journal volume SJVOL to which the volume ID of “102” is assigned in the secondary site **3** sequentially.

[0137] Furthermore, FIG. **12** also illustrates that the path ID of the remote copy path **7** connecting from the primary volume PVOL to the secondary volume SVOL of the volume pair is “0” and the current state is normal (“Normal”).

[0138] In addition, in FIG. **12**, the secondary volume SVOL of the volume pair to which the volume pair ID of “0” is assigned and the secondary volume SVOL of the volume pair to which the volume pair ID of “1” is assigned are redundant logical volumes since the primary volumes PVOL of these volume pairs are the same (the volume ID of the primary volume PVOL is “1”), and the secondary volume SVOL of the volume pair to which the volume pair ID of “O” is assigned is the active secondary volume SVOL (the attribute of the secondary volume SVOL is “Active”).

[0139] The path management table **45** is a table used to manage all the remote copy paths **7** set in the storage system **1**. In the following description, it is assumed that all remote copy paths are duplicated. As illustrated in FIG. **13**, the path management table **45** includes a path ID field **45A**, a protocol information field **45B**, an IP address field **45C**, an access policy field **45D**, and a priority path field **45E**. In the path management table **45**, one record corresponds to one remote copy path **7** set in the storage system **1**.

[0140] The path ID field **45A** stores the path ID of the corresponding remote copy path **7**, and the protocol information field **45B** stores the protocol with which the remote copy path **7** conforms.

[0141] The IP address field **45C** is divided into a first path field **45CA** and a second path field **45CB** so as to correspond to each of two paths (hereinafter referred to as first and second paths) constituting the corresponding remote copy path **7**. Then, in the first path field **45CA** and the second path field **45CB**, the IP address (the IP address represented by “P” in FIG. **13**) of the port **23** of the first storage node **11** on the primary site **2** side to which the corresponding first or second path is connected and the IP address (the IP address represented by “S” in the figure) of the port **23** of the second storage node **30** on the secondary site **3** side are stored.

[0142] Furthermore, in the access policy field **45D**, information indicating how to use the first and second paths constituting the corresponding remote copy path **7** is stored. As a method of using the first and second paths in this case, for example, there are “symmetric” in which both the first and second paths are used while being switched by a method such as round robin “asymmetric” in which only one of the first or second paths is used, and the like. Furthermore, in the priority path field **45E**, the IP address of the port **23** on the first storage node **11** side of the first or second path and the IP address of the port **23** of the second storage node **30** used when the access policy is “asymmetric” are stored.

[0143] Therefore, in the case of the example of FIG. **13**, for example, it is illustrated that the remote copy path **7** to which the path ID “O” is assigned includes a first path connecting the port **23** to which the IP address “172.12.16.202” is assigned of a certain first storage node **11** installed in the primary site **2** and the port **23** to which the IP address “172.12.16.203” is assigned of a certain second storage node **30** installed in the secondary site **3**, and a second path connecting the port **23**

to which the IP address “172.12.16.302” is assigned of the certain first storage node **11** and the port **23** to which the IP address “172.12.16.303” is assigned of the second storage node. FIG. **13** also illustrates that the remote copy path **7** is a path conforming to “iSCSI”, the access policy is “asymmetric”, and the first path is a priority path.

[0144] Note that since the first and second storage nodes **11** and **30** recognize the communication destination with reference to the path management table **45** held therein, the setting of the remote copy path **7** can be performed by registering necessary information in the path management table **45** (creating a record corresponding to the remote copy path **7**), and the deletion of the remote copy path **7** can be performed by deleting the information of the remote copy path **7** stored in the path management table **45** (deleting the record corresponding to the remote copy path **7**).

[0145] On the other hand, the I/O processing unit **50** is a part of the storage control software **35** mounted on the first and second storage nodes **11** and **30**, and has a function of executing the I/O process corresponding to the I/O request in a case where the I/O request (read request or write request) is provided from the host device **10** (FIG. **1**) for the first storage node **11**, and in a case where the write request of data of the remote copy target is provided from the first storage node **11** for the second storage node **30**.

[0146] In practice, when a data read request is given from the host device **10** (FIG. **1**), the I/O processing unit **50** of the first storage node **11** reads the data from the position where the data to be read designated in the read request in the primary volume PVOL where the data to be read designated in the read request is stored, and transmits the data to the host device **10**.

[0147] In addition, in a case where a data write request is given from the host device **10**, the I/O processing unit **50** of the first storage node **11** writes the data to be written at the position designated in the write request in the primary volume PVOL to which the data to be written designated in the write request is to be written, generates the journal data JNL of the data, stores the generated journal data JNL in the corresponding primary journal volume PJVOL, and further transfers the journal data JNL to the corresponding second storage node **30**.

[0148] Note that since the processing contents (I/O processing) of the I/O processing unit **50** of the second storage node **30** have already been described, the description thereof will be omitted here.

[0149] The node increasing processing unit **51** is a part of the storage control software **35** implemented in the first and second storage nodes **11** and **30** for the first and second storage nodes **11** and **30**, and is a part of the primary site management software **26** (FIG. **1**) or the secondary site management software **34** (FIG. **1**) implemented in the primary site management device **12** and the secondary site management device **31** for the primary site management device **12** and the secondary site management device **31**. When the second storage node **30** is increased in the secondary site **3**, the node increasing processing unit **51** executes a necessary process among a series of processes performed in the storage system **1** as described above.

[0150] The node decreasing processing unit **52** is a part of the storage control software **35** implemented in the first and second storage nodes **11** and **30** for the first and second storage nodes **11** and **30**, and is a part of the primary site management software **26** (FIG. **1**) or the secondary site management software **34** (FIG. **1**) implemented in the primary site management device **12** and the secondary site management device **31** for the primary site management device **12** and the secondary site management device **31**. When the second storage node **30** is decreased from the secondary site **3**, the node decreasing processing unit **52** executes a necessary process among a series of processes performed in the storage system **1** as described above.

(3) VARIOUS PROCESSES RELATED TO PATH MANAGEMENT FUNCTION OF PRESENT EMBODIMENT

[0151] Next, specific processing contents of various processes executed in relation to the path management function of the present embodiment will be described. Note that, in the following description, a processing subject of various processes will be described as a “program”, but in practice, it goes without saying that the processor **24** (FIG. **1**) in the second storage node **30** or a

processor (not illustrated) of the primary site management device **12** or the secondary site management device **31** executes the process based on the program.

(3-1) Secondary Site Management Software Side Storage Node Increasing Process

[0152] FIG. **14** illustrates a flow of a series of processes (hereinafter referred to as secondary site management software side storage node increasing process) executed by the secondary site management software **34** of the secondary site management device **31** when a new second storage node **30** is increased to the cluster **33** in the secondary site **3**.

[0153] Note that in the following description, it is assumed that the active storage control software **35** and the active secondary volume SVOL for which the active storage control software **35** has the ownership are already arranged in the increased storage node **30** by a user operation in advance or the like, and necessary information regarding the increased storage node **30**, the active storage control software **35**, and the active secondary volume SVOL is already registered in the system configuration management table **40**, the storage device management table **41**, the port management table **42**, and the volume management table **43**.

[0154] When information indicating that the second storage node **30** is increased in the cluster **33** is provided from the user, for example, by the user performing a predetermined operation on the secondary site management device **31**, the secondary site management software **34** starts the secondary site management software side storage node increasing process illustrated in FIG. **14**.

[0155] Then, the secondary site management software **34** first registers the increased storage node **30** in the cluster **33** of the increased destination (S1). Specifically, the secondary site management software **34** registers and manages information on which second storage node **30** belongs to which cluster **33** in a management table (hereinafter referred to as a cluster management table) not illustrated, and registers information indicating that the second storage node **30** is increased to the cluster **33** in the cluster management table.

[0156] Subsequently, the secondary site management software **34** selects the second storage node (hereinafter referred to as a rearrangement source storage node) **30** of a rearrangement source of the standby storage control software **35** (and the standby secondary volume SVOL) to be rearranged in the increased storage node **30** (S2). For example, the secondary site management software **34** refers to the volume management table **43** held therein, determines the standby storage control software **35** having the smallest number of managed secondary volumes SVOL as the standby storage control software **35** to be rearranged, and selects the second storage node **30** in which the standby storage control software **35** is arranged as the rearrangement source storage node **30**.

[0157] At this time, the secondary site management software **34** selects the rearrangement source storage node **30** to be selected by the number corresponding to the redundancy of the secondary volume SVOL (the number of storage control software **35** constituting the redundancy group). For example, the secondary site management software **34** selects N rearrangement source storage nodes **30** when the redundancy of the secondary volume SVOL is “N”.

[0158] Next, the secondary site management software **34** rearranges the standby storage control software **35** to be rearranged in each rearrangement source storage node **30** selected as described above in the increased storage node **30**, and moves all the standby secondary volumes SVOL in which the rearranged standby storage control software **35** has the ownership in the second storage node **30** of the rearrangement source and the configuration information thereof to the increased storage node **30** (S3).

[0159] Specifically, the secondary site management software **34** first gives an instruction to each of the second storage nodes **30** in which each of the standby storage control software **35** to be rearranged is arranged and the increased storage node **30** so as to copy all of the data of each of the standby storage control software **35** to be rearranged, the volume data of all the standby secondary volumes SVOL for which the standby storage control software **35** has the ownership, and the configuration information of the standby secondary volumes SVOL to the increased storage node **30**.

[0160] In addition, the secondary site management software **34** rewrites the value of the node ID field **43F** (FIG. **11**) of the record corresponding to each active secondary volume SVOL moved to the increased storage node **30** as described above in the volume management table **43** (FIG. **11**) held therein to the node ID of the increased storage node **30**.

[0161] Then, the secondary site management software **34** transmits differential data indicating the difference of before and after the update in the volume management table **43** updated in this manner to each of the second storage nodes **30** in the secondary site **3**. As a result, the contents of the volume management table **43** held by each of the second storage nodes **30** are updated based on the differential data similarly to the volume management table **43** held by the secondary site management software **31**.

[0162] Note that the secondary site management software **34** also transmits the differential data to the primary site management software **26** of the primary site **2**. Then, the primary site management software **26** that has received the differential data updates the volume management table **43** held therein based on the differential data similarly to the volume management table **43** held by the secondary site management software **31**.

[0163] Furthermore, the secondary site management software **34** transmits differential data indicating the difference of before and after the update in the volume management table **43** updated in this manner to each of the first storage nodes **11** in the primary site **2**. As a result, the contents of the volume management table **43** held by each of the first storage nodes **11** are updated based on the differential data similarly to the volume management table **43** held by the primary site management software **26**.

[0164] Subsequently, the secondary site management software **34** gives an instruction to each second storage node **30** of a rearrangement source of each rearranged standby storage control software **35** to create the standby storage control software **35** configuring the redundancy group together with the active storage control software **35** created in the increased storage node **30** (S4). As a result, the standby storage control software **35** configuring the same redundancy group together with the active storage control software **35** created in the increased storage node **30** is created in each of the second storage nodes **30**.

[0165] Thereafter, the secondary site management software **34** transmits to the pair management software **6** of the pair management device **5**, the storage control software ID of each standby storage control software **35** rearranged in the increased storage node **30** as described above, the volume IDs of all the standby secondary volumes SVOL for which the standby storage control software **35** has the ownership, the node ID of the rearrangement source storage node **30** of the standby storage control software **35**, and the node ID of the second storage node (increased storage node) of the rearrangement destination as the rearrangement information (S5).

[0166] The secondary site management software **34** thereafter ends the secondary site management software side storage node increasing process.

(3-2) Process of Pair Management Software at Time of Storage Node Increase

[0167] FIG. **15** illustrates a flow of a series of processes (hereinafter referred to as a pair management software side storage node increasing process) executed by the pair management software **6** of the pair management device **5** that has received the rearrangement information described above.

[0168] When the rearrangement information is transmitted, the pair management software **6** starts the pair management software side storage node increasing process illustrated in FIG. **15**, and first acquires (receives) the transmitted rearrangement information (S10).

[0169] Subsequently, the pair management software **6** selects one standby secondary volume SVOL unprocessed after step S12 from among the standby secondary volumes SVOL moved to the increased storage node **30** recognized based on the acquired rearrangement information (S11).

[0170] Next, the pair management software **6** collects, from the primary site management device **12**, information necessary for setting the remote copy path **7** for performing remote copy between

the standby secondary volume (hereinafter referred to as a selected standby secondary volume) SVOL selected in step **S11** and the primary volume PVOL corresponding to the selected standby secondary volume SVOL between the first storage node (hereinafter referred to as a specific storage node in the description of FIG. 15) **11** in which the primary volume PVOL is created and the increased storage node **30** (**S12**).

[0171] Specifically, the pair management software **6** designates the volume ID of the selected standby secondary volume SVOL, and inquires the primary site management software **26** of the primary site management device **12** about the volume ID of the primary volume PVOL corresponding to the selected standby secondary volume SVOL and the storage control software ID of the storage control software **35** having the ownership of the primary volume PVOL. In addition, the pair management software **6** inquires the primary site management software **26** of the primary site management device **12** about the IP address of each port **23** included in the specific storage node **11** in which the storage control software **35** is arranged and the IP address of each port **23** included in the increased storage node **30**.

[0172] The primary site management software **26** that has received this inquiry acquires the volume ID of the primary volume PVOL corresponding to the selected standby secondary volume SVOL from the volume pair management table **44** (FIG. 9) held therein. In addition, the primary site management software **26** acquires the storage control software ID of the storage control software **35** having the ownership of the primary volume PVOL and the node ID of the specific storage node **11** in which the primary volume PVOL is created from the volume management table **43** held therein.

[0173] Furthermore, the primary site management software **26** acquires the port ID of each port **23** included in the specific storage node **11** and the port ID of each port **23** included in the increased storage node **30** from the system configuration table **40** (FIG. 5) held therein. Then, the primary site management software **26** acquires the IP addresses of each of the ports **23** of the specific storage node **11** and the increased storage node **30**, whose port IDs have been acquired as described above, from the port management table **42** (FIG. 7) held therein. Then, the primary site management software **26** transmits these pieces of information acquired in this way to the pair management software **6**.

[0174] Subsequently, the pair management software **6** gives an instruction to the primary site management software **26** and the secondary site management software **34** to create the remote copy path **7** connecting the primary volume PVOL, whose volume ID has been acquired in step **S12**, and the selected standby secondary volume SVOL between the specific storage node **11** and the increased storage node **30** (**S13**).

[0175] Specifically, the pair management software **6** creates two combinations of the IP address of each port **23** of the specific storage node **11** acquired in step **S12** and the IP address of each port **23** of the increased storage node **30**, and gives an instruction to each of the primary site management software **26** and the secondary site management software **34** to create a path connecting these IP addresses for each combination.

[0176] Thus, the primary site management software **26** and the secondary site management software **34** that have received this instruction store the combinations of the IP addresses as the combinations of the IP addresses on both end sides of the first path and the second path in the first and second path fields **45CA** and **45CB** of the unused record of the path management table **45** (FIG. 13) held therein, respectively, and store necessary information (e.g., information defined by default) in the protocol information field **45B**, the access policy field **45D**, and the priority path field **45E** of the record. The primary site management software **26** and the secondary site management software **34** store the same path ID assigned to the remote copy path **7** in the path ID field **45A** of the path management table **45**.

[0177] Furthermore, the primary site management software **26** and the secondary site management software **34** update the value of the path ID field **44F** of the record in which the selected standby

secondary volume SVOL in the volume pair management table **44** held therein is stored in the secondary volume ID field **44E** (FIG. **12**) to the same path ID assigned to the remote copy path **7**. [0178] Then, the primary site management software **26** transmits differential data of before and after the update, as described above, of the path management table **45** and the volume pair management table **44** held therein to each of the first storage nodes **11** in the primary site **2**, thereby similarly updating the path management table **45** and the volume pair management table **44** held by each of the first storage nodes **11**. Then, the secondary site management software **34** transmits differential data of before and after the update, as described above, of the path management table **45** and the volume pair management table **44** held therein to each of the second storage nodes **30** in the secondary site **3**, thereby similarly updating the path management table **45** and the volume pair management table **44** held by each of the second storage nodes **30**.

[0179] As described above, the remote path **7** is set between the specific storage node **11** and the increased storage node **30**. Note that the primary site management software **26** and the secondary site management software **34** may further cooperate to create an iSCSI initiator associated with the primary volume PVOL corresponding to the selected standby secondary volume SVOL in the specific storage node **11**, create an iSCSI target associated with the selected standby secondary volume SVOL in the increased storage node **30**, and perform processes until a session between the iSCSI initiator and the iSCSI target is established.

[0180] Next, the pair management software **6** collects, from the primary site management device **12**, information necessary for deleting the remote copy path **7** connecting the second storage node (hereinafter referred to as a movement source storage node) **30** of the movement source of the selected standby secondary volume SVOL and the specific storage node **11** (S14). Specifically, the pair management software **6** extracts the node ID of the movement source storage node **30** from the rearrangement information notified from the secondary site management software **34**, and inquires the primary site management software **26** about the IP address of each port **23** included in the movement source storage node **30** to which the node ID is assigned.

[0181] The primary site management software **26** that has received this inquiry acquires the port ID of each port **23** included in the movement source storage node **30** from the record corresponding to the movement source storage node **30** in the system configuration table **40** (FIG. **5**) held therein. Then, the primary site management software **26** acquires the IP address of each port **23** of the movement source storage node **30** that has acquired the port ID in the above manner from the port management table **42** (FIG. **7**) held therein. The primary site management software **26** then transmits the acquired IP address of each port **23** of the movement source storage node **30** to the pair management software **6**.

[0182] The pair management software **26** gives an instruction to the primary site management software **26** and the secondary site management software **34** to delete the remote copy path **7** connecting the specific storage node **11** and the movement source storage node **30** based on the IP address of each port **23** included in the movement source storage node **30** acquired in this manner and the IP address of each port **23** included in the specific storage node **11** acquired in step S12 (S15). Note that this instruction includes information on the IP address of each port **23** included in the specific storage node **11** and information on the IP address of each port **23** included in the movement source storage node **30**.

[0183] Thus, the primary site management software **26** and the secondary site management software **34** that have received this instruction specify a record in which at least one of the combinations of the IP address of each port **23** included in the specific storage node **11** and the IP address of each port **23** included in the movement source storage node **30** is stored in the first or second path fields **45CA** and **45CB** from the records of the path management table **45** held therein, and delete the record from the path management table **45**.

[0184] In addition, the primary site management software **26** transmits differential data of before and after the update as described above of the path management table **45** held therein to each of the

first storage nodes **11** in the primary site **2**, thereby similarly updating the path management table **45** held by each of the first storage nodes **11**. Furthermore, the secondary site management software **34** also transmits differential data of before and after the update as described above of the path management table **45** held therein to each of the second storage nodes **30** in the secondary site **3**, thereby similarly updating the path management table **45** held by each of the second storage nodes **30**. As described above, the remote path **7** set between the specific storage node **11** and the movement source storage node **30** is deleted.

[0185] Thereafter, the pair management software **26** determines whether or not the processes after step **S12** have been executed for all the standby secondary volumes SVOL moved to the increased storage node **30** recognized from the rearrangement information notified from the secondary site management software **34** (**S16**). Then, when a negative result is obtained in this determination, the pair management software **26** returns to step **S11**, and thereafter, repeats the processes of steps **S11** to **S16** while sequentially switching the standby secondary volume SVOL selected in step **S11** to another corresponding standby secondary volume SVOL which is not processed after step **S12**.

[0186] According to this repetition process, the remote copy path **7** connecting between each standby secondary volume SVOL moved to the increased storage node **30** and the primary volume PVOL corresponding to the standby secondary volume SVOL is sequentially set between the corresponding first storage node **11** and the increased storage node **30**, and the remote copy path **7** set between the movement source storage node **30** and the specific storage node **11** of the standby secondary volume SVOL is sequentially deleted.

[0187] When obtaining a positive result in step **S16** by finishing the execution of the processes of step **S12** to step **S15** for each standby secondary volume SVOL moved to the increased storage node **30** in due course, the pair management software **6** terminates the pair management software side storage node increasing process.

(3-3) Process of Secondary Site Management Software at Time of Storage Node Decrease

[0188] FIG. **16** illustrates a flow of a series of processes (hereinafter referred to as secondary site management software side storage node decreasing process) executed by the secondary site management software **34** of the secondary site management device **31** when decreasing the second storage node **30** from the cluster **33** in the secondary site **3**.

[0189] When information indicating that the specific second storage node **30** is to be decreased from the cluster **33** is given from the user, for example, by the user performing a predetermined operation on the secondary site management device **31**, the secondary site management software **34** starts the secondary site management software side storage node decreasing process illustrated in FIG. **13**.

[0190] Then, the secondary site management software **34** first selects the active storage control software **35** to which the ownership of all the active secondary volumes SVOL owned by the active storage control software **35** arranged in the decreased storage node **30** is transferred from the active storage control software **35** arranged in the second storage node **30** other than the decreased storage node **30** in the same cluster **33** as the second storage node (decreased storage node) **30** to be decreased (**S20**).

[0191] For example, the secondary site management software **34** selects the active storage control software **35** so that the loads of the second storage nodes **30** other than the decreased storage node **30** in the cluster **33** to which the decreased storage node **30** belongs are balanced. Hereinafter, the active storage control software **35** selected at this time is referred to as transfer destination active storage control software **35**.

[0192] Subsequently, the secondary site management software **34** moves all the active secondary volumes SVOL owned by the active storage control software **35** arranged in the decreased storage node **30** to the second storage node **30** in which the transfer destination active storage control software **35** selected in step **S20** is arranged, and moves the ownership of each of the moved active secondary volumes SVOL to the transfer destination active storage control software **35** (**S21**).

[0193] Specifically, the secondary site management software **34** gives an instruction to the second storage node **30** and the decreased storage node **30** to copy the data of the active secondary volume SVOL and the configuration information of the active secondary volume SVOL to the second storage node **30** in which the transfer destination active storage control software **35** is arranged for each active secondary volume SVOL owned by the active storage control software **35** arranged in the secondary storage node **30**.

[0194] In addition, the secondary site management software **34** rewrites the storage control software ID stored in the owner ID field **43B** (FIG. **11**) of the record corresponding to each active secondary volume SVOL moved to the second storage node **30** in which the transfer destination active storage control software **35** is arranged as described above in the volume management table **43** (FIG. **11**) held therein to the storage control software ID of the transfer destination active storage control software **35**. In addition, the secondary site management software **34** rewrites the value of the node ID field **43F** of the record to the node ID of the second storage node **30** in which the transfer destination active storage control software **35** is arranged.

[0195] Then, the secondary site management software **34** transmits differential data indicating the difference of before and after the update in the volume management table **43** updated in this manner to each of the second storage nodes **30** in the secondary site **3**. As a result, the contents of the volume management table **43** held by each of the second storage nodes **30** are updated based on the differential data similarly to the volume management table **43** held by the secondary site management software **31**. As a result, the ownership of each active secondary volume SVOL is transferred to the movement destination active storage control software **35**.

[0196] Note that the secondary site management software **34** also transmits the differential data to the primary site management software **26** of the primary site **2**. Then, the primary site management software **26** that has received the differential data updates the volume management table **43** held therein based on the differential data similarly to the volume management table **43** held by the secondary site management software **31**.

[0197] Furthermore, the secondary site management software **34** transmits differential data indicating the difference of before and after the update in the volume management table **43** updated in this manner to each of the first storage nodes **11** in the primary site **2**. As a result, the contents of the volume management table **43** held by each of the first storage nodes **11** are updated based on the differential data similarly to the volume management table **43** held by the primary site management software **26**.

[0198] Subsequently, the secondary site management software **34** moves each standby secondary volume SVOL owned by the standby storage control software **35** configuring the same redundancy group as the active storage control software **35** arranged in the decreased storage node **30** and its configuration information to the second storage node **30** in which the standby storage control software **35** configuring the same redundancy group as the transfer destination active storage control software **35** is arranged (S22).

[0199] Specifically, the secondary site management software **34** copies each standby secondary volume SVOL and the configuration information thereof to the second storage node **30** in which the standby storage control software **35** configuring same redundancy group as the transfer destination active storage control software **35** is arranged, similarly to step S21.

[0200] Thereafter, the secondary site management software **34** moves the ownership of the standby secondary volume SVOL to the standby storage control software **35** (corresponding standby storage control software **35** configuring the same redundancy group as the transfer destination active storage control software **35**) in the second storage node **30** of the movement destination in the same manner as described above. Furthermore, the secondary site management software **34** updates the volume management table **43** held by each of the secondary site management device **31**, the second storage node **30**, the primary site management device **12**, and the first storage nodes **11** in response to the update similarly to step S21.

[0201] Next, the secondary site management software **34** deletes the active storage control software **35** arranged in the decreased storage node **30** from the decreased storage node **30**, and deletes the standby storage control software **35** configuring the same redundancy group as the active storage control software **35** from the second storage node **30** in which the standby storage control software **35** is arranged (S23).

[0202] Specifically, the secondary site management software **34** gives an instruction to delete the data of the active storage control software **35** with respect to the decreased storage node **30**, and gives an instruction to the second storage node **30** in which the standby storage control software **35** is arranged to delete the data of the standby storage control software **35** constituting the same redundancy group as the active storage control software **35**.

[0203] Subsequently, the secondary site management software **34** moves all the standby storage control software **35** arranged in the decreased storage node **30**, the standby secondary volume SVOL owned by the standby storage control software **35**, and the configuration information of the standby secondary volume SVOL to any second storage node **30** other than the decreased storage node **30** in the same cluster **33** (S24).

[0204] Specifically, the secondary site management software **34** first determines the second storage node **30** to be the rearrangement destination of each piece of standby storage control software **35** arranged in the decreased storage node **30**. The rearrangement destination determines to balance the loads of each of the second storage nodes **30** in the same cluster **33**. Then, the secondary site management software **34** gives an instruction to copy the data of the standby storage control software **35**, the standby secondary volume SVOL, and the configuration information thereof to each of the determined second storage nodes **30** and the decreased storage node **30**.

[0205] Thereafter, the secondary site management software **34** rewrites the value stored in the node ID field **43F** corresponding to the standby secondary volume SVOL in the volume management table **43** held by each of the secondary site management device **31**, the second storage node **30**, the primary site management device **12**, and the first storage nodes **11** to the node ID of the second storage node **30** of the movement destination of the standby secondary volume SVOL, similarly to step S21, according to the movement of the standby secondary volume SVOL.

[0206] Then, the secondary site management software **34** transmits, to the pair management software **6** of the pair management device **5**, the storage control software ID of the storage control software **35** rearranged (including the transfer of the ownership) from the decreased storage node **30** to another second storage node **30** as described above, the volume ID of each secondary volume SVOL moved accordingly from the decreased storage node **30** to another second storage node **30** accompanying therewith, the node ID of the second storage node **30** of the rearrangement destination of each rearranged storage control software **35**, and the node ID of the decreased storage node **30** as the rearrangement information (S25).

[0207] The secondary site management software **34** thereafter ends the secondary site management software side storage node decreasing process.

(3-4) Process of Pair Management Software at Time of Storage Node Decrease

[0208] FIG. 17A and FIG. 17B illustrate a flow of a series of processes (hereinafter referred to as a pair management software side storage node decreasing process) executed by the pair management software **6** of the pair management device **5** that has received the rearrangement information described above.

[0209] When such rearrangement information is transmitted, the pair management software **6** starts the pair management software side storage node decreasing process illustrated in FIG. 17A and [0210] FIG. 17B, and first receives and acquires the transmitted rearrangement information (S30).

[0211] Subsequently, the pair management software **6** selects one active secondary volume SVOL which is not processed after step S32 from among the active secondary volumes SVOL recognized based on the acquired rearrangement information and moved from the decreased storage node **30** to another second storage node **30** (S31).

[0212] Next, the pair management software **6** collects, from the primary site management device **12**, information necessary for setting the remote copy path **7** for performing remote copy between the active secondary volume (hereinafter referred to as a selected active secondary volume) SVOL selected in step **S31** and the primary volume PVOL corresponding to the selected active secondary volume SVOL between the first storage node (hereinafter referred to as an active specific storage node) **11** in which the primary volume PVOL is created and the second storage node (hereinafter referred to as an active movement destination storage node) **30** of the movement destination of the selected active secondary volume (**S32**).

[0213] Specifically, the pair management software **6** designates the volume ID of the selected active secondary volume SVOL, and inquires the primary site management software **26** of the primary site management device **12** about the volume ID of the primary volume PVOL corresponding to the selected active secondary volume SVOL and the storage control software ID of the storage control software **35** having the ownership of the primary volume PVOL. In addition, the pair management software **6** inquires the primary site management software **26** of the primary site management device **12** about the IP address of each port **23** included in the active specific storage node **11** in which the storage control software **35** is arranged and the IP address of each port **23** included in the active movement destination storage node **30**.

[0214] The primary site management software **26** that has received this inquiry acquires the volume ID of the primary volume PVOL corresponding to the selected active secondary volume SVOL from the volume pair management table **44** (FIG. **9**) held therein. In addition, the primary site management software **26** acquires the storage control software ID of the storage control software **35** having the ownership of the primary volume PVOL and the node ID of the active specific storage node **11** in which the primary volume PVOL is created from the volume management table **43** held therein.

[0215] Furthermore, the primary site management software **26** acquires the port ID of each port **23** included in the active specific storage node **11** and the port ID of each port **23** included in the active movement destination storage node **30** from the system configuration table **40** (FIG. **5**) held therein. Then, the primary site management software **26** acquires the IP addresses of the ports **23** of the active specific storage node **11** and the active movement destination storage node **30**, for which the port IDs have been acquired as described above, from the port management table **42** (FIG. **7**) held therein. Then, the primary site management software **26** transmits these pieces of information acquired in this way to the pair management software **6**.

[0216] Subsequently, the pair management software **6** gives an instruction to the primary site management software **26** and the secondary site management software **34** to create the remote copy path **7** connecting the primary volume PVOL of which the volume ID is acquired in step **S32** and the selected active secondary volume SVOL between the active specific storage node **11** and the active movement destination storage node **30** (**S33**).

[0217] Specifically, the pair management software **6** creates two combinations of the IP address of each port **23** of the active specific storage node **11** acquired in step **S32** and the IP address of each port **23** of the active movement destination storage node **30**, and gives an instruction to each of the primary site management software **26** and the secondary site management software **34** to create a path connecting these IP addresses for each combination.

[0218] Thus, the primary site management software **26** and the secondary site management software **34** that have received this instruction store the combinations of the IP addresses as the combinations of the IP addresses on both end sides of the first path and the second path in the first and second path fields **45CA** and **45CB** of the unused record of the path management table **45** (FIG. **13**) held therein, respectively, and store necessary information (e.g., information defined by default) in the protocol information field **45B**, the access policy field **45D**, and the priority path field **45E** of the record. The primary site management software **26** and the secondary site management software **34** store the same path ID assigned to the remote copy path **7** in the path ID

field **45A** of the path management table **45**.

[0219] Furthermore, the primary site management software **26** and the secondary site management software **34** update the value of the path ID field **44F** of the record in which the selected active secondary volume SVOL in the volume pair management table **44** held therein is stored in the secondary volume ID field **44E** (FIG. **12**) to the same path ID assigned to the remote copy path **7**.

[0220] Then, the primary site management software **26** transmits differential data of before and after the update, as described above, of the path management table **45** and the volume pair management table **44** held therein to each of the first storage nodes **11** in the primary site **2**, thereby similarly updating the path management table **45** and the volume pair management table **44** held by each of the first storage nodes **11**. Then, the secondary site management software **34** transmits differential data of before and after the update, as described above, of the path management table **45** and the volume pair management table **44** held therein to each of the second storage nodes **30** in the secondary site **3**, thereby similarly updating the path management table **45** and the volume pair management table **44** held by each of the second storage nodes **30**.

[0221] As described above, the remote path **7** is set between the active specific storage node **11** and the active movement destination storage node **30**. The primary site management software **26** and the secondary site management software **34** may further cooperate to create an iSCSI initiator associated with the primary volume PVOL corresponding to the selected active secondary volume SVOL in the active specific storage node **11**, create an iSCSI target associated with the selected active secondary volume SVOL in the active movement destination storage node **30**, and perform processes until a session between the iSCSI initiator and the iSCSI target is established.

[0222] Next, the pair management software **6** collects, from the primary site management devices **12**, information necessary for deleting the remote copy path **7** connecting the decreased storage node **30** of the movement source of the selected active secondary volume SVOL and the active specific storage node **11** (S34). Specifically, the pair management software **6** extracts the node ID of the decreased source storage node **30** from the rearrangement information notified from the secondary site management software **34**, and inquires the primary site management software **26** about the IP address of each port **23** included in the decreased storage node **30** to which the node ID is assigned.

[0223] The primary site management software **26** that has received this inquiry acquires the port ID of each port **23** included in the decreased storage node **30** from the record corresponding to the decreased storage node **30** in the system configuration table **40** (FIG. **5**) held therein. Then, the primary site management software **26** acquires the IP address of each port **23** of the decreased storage node **30** that has acquired the port ID in the above manner from the port management table **42** (FIG. **7**) held therein. The primary site management software **26** then transmits the acquired IP address of each port **23** of the acquired decreased storage node **30** to the pair management software **6**.

[0224] Then, the pair management software **6** gives an instruction to the primary site management software **26** and the secondary site management software **34** to delete the remote copy path **7** connecting the active specific storage node **11** and the decreased storage node **30** based on the IP address of each port **23** included in the decreased storage node **30** acquired in the above manner and the IP address of each port **23** included in the active specific storage node **11** acquired in step S32 (S35). Note that this instruction includes information on the IP address of each port **23** included in the active specific storage node **11** and information on the IP address of each port **23** included in the decreased storage node **30**.

[0225] Thus, the primary site management software **26** and the secondary site management software **34** that have received this instruction specify a record in which at least one of the combinations of the IP address of each port **23** included in the active specific storage node **11** and the IP address of each port **23** included in the decreased storage node **30** is stored in the first or second path fields **45CA** and **45CB** from the records of the path management table **45** held therein,

and delete the record from the path management table **45**.

[0226] In addition, the primary site management software **26** transmits differential data of before and after the update as described above of the path management table **45** held therein to each of the first storage nodes **11** in the primary site **2**, thereby similarly updating the path management table **45** held by each of the first storage nodes **11**. Furthermore, the secondary site management software **34** also transmits differential data of before and after the update as described above of the path management table **45** held therein to each of the second storage nodes **30** in the secondary site **3**, thereby similarly updating the path management table **45** held by each of the second storage nodes **30**. As described above, the remote path **7** set between the active specific storage node **11** and the decreased storage node **30** is deleted.

[0227] Thereafter, the pair management software **26** determines whether or not the processes after step **S32** have been finished execution for all the active secondary volumes SVOL moved from the decreased storage node **30** from the rearrangement information notified from the secondary site management software **34** to the other second storage nodes **30** (**S36**). Then, when a negative result is obtained in this determination, the pair management software **26** returns to step **S31**, and thereafter, repeats the processes of steps **S31** to **S36** while sequentially switching the active secondary volume SVOL selected in step **S31** to another corresponding active secondary volume SVOL which is not processed after step **S32**.

[0228] By this repetition process, the remote copy path **7** connecting between each active secondary volume SVOL moved from the decreased storage node **30** to another second storage node **30** and the primary volume PVOL corresponding to the active secondary volume SVOL is sequentially set between the corresponding active specific storage node **11** and the active movement destination storage node **30**, and the original remote copy path **7** set between the decreased storage node **30** that is the movement source of the active secondary volume SVOL and the active specific storage node **11** is sequentially deleted.

[0229] Then, when obtaining a positive result in step **S36** by finishing the execution of the processes of step **S32** to step **S35** for each standby secondary volume SVOL moved to the increased storage node **30** in due course, the pair management software **6** selects one standby secondary volume SVOL that is not processed after step **S38** from among the standby secondary volumes SVOL managed by each standby storage control software **35** arranged in the decreased storage node **30** moved from the decreased storage node **30** to the other second storage node **30** (**S37**).

[0230] Next, the pair management software collects, from the primary site management device **12**, information necessary for setting the remote copy path **7** for performing remote copy between the active secondary volume (hereinafter referred to as a selected standby secondary volume) SVOL selected in step **S37** and the primary volume PVOL corresponding to the selected standby secondary volume SVOL between the first storage node (hereinafter referred to as a standby specific storage node) **11** in which the primary volume PVOL is created and the second storage node (hereinafter referred to as a standby movement destination storage node) **30** of the movement destination of the selected standby secondary volume (**S38**).

[0231] Specifically, the pair management software **6** designates the volume ID of the selected standby secondary volume SVOL, and inquires the primary site management software **26** of the primary site management device **12** about the volume ID of the primary volume PVOL corresponding to the selected standby secondary volume SVOL and the storage control software ID of the storage control software **35** having the ownership of the primary volume PVOL. In addition, the pair management software **6** inquires the primary site management software **26** of the primary site management device **12** about the IP address of each port **23** included in the standby specific storage node **11** in which the storage control software **35** is arranged and the IP address of each port **23** included in the standby movement destination storage node **30**.

[0232] The primary site management software **26** that has received this inquiry acquires the volume

ID of the primary volume PVOL corresponding to the selected standby secondary volume SVOL from the volume pair management table **44** (FIG. **9**) held therein. In addition, the primary site management software **26** acquires the storage control software ID of the storage control software **35** having the ownership of the primary volume PVOL and the node ID of the standby specific storage node **11** in which the primary volume PVOL is created from the volume management table **43** held therein.

[0233] Furthermore, the primary site management software **26** acquires the port ID of each port **23** included in the standby specific storage node **11** and the port ID of each port **23** included in the standby movement destination storage node **30** from the system configuration table **40** (FIG. **5**) held therein. Then, the primary site management software **26** acquires the IP addresses of the ports **23** of the standby specific storage node **11** and the standby movement destination storage node **30**, for which the port IDs have been acquired as described above, from the port management table **42** (FIG. **7**) held therein. Then, the primary site management software **26** transmits these pieces of information acquired in this way to the pair management software **6**.

[0234] Subsequently, the pair management software **6** gives an instruction to the primary site management software **26** and the secondary site management software **34** to create the remote copy path **7** connecting the primary volume PVOL of which the volume ID is acquired in step S38 and the selected standby secondary volume SVOL between the standby specific storage node **11** and the standby movement destination storage node **30** (S39).

[0235] Specifically, the pair management software **6** creates two combinations of the IP address of each port **23** of the standby specific storage node **11** acquired in step S38 and the IP address of each port **23** of the standby movement destination storage node **30**, and gives an instruction to each of the primary site management software **26** and the secondary site management software **34** to create a path connecting these IP addresses for each combination.

[0236] Thus, the primary site management software **26** and the secondary site management software **34** that have received this instruction store the combinations of the IP addresses as the combinations of the IP addresses on both end sides of the first path and the second path in the first and second path fields **45CA** and **45CB** of the unused record of the path management table **45** (FIG. **13**) held therein, respectively, and store necessary information (e.g., information defined by default) in the protocol information field **45B**, the access policy field **45D**, and the priority path field **45E** of the record. The primary site management software **26** and the secondary site management software **34** store the same path ID assigned to the remote copy path **7** in the path ID field **45A** of the path management table **45**.

[0237] Furthermore, the primary site management software **26** and the secondary site management software **34** update the value of the path ID field **44F** of the record in which the selected standby secondary volume SVOL in the volume pair management table **44** held therein is stored in the secondary volume ID field **44E** (FIG. **12**) to the same path ID assigned to the remote copy path **7**.

[0238] Then, the primary site management software **26** transmits differential data of before and after the update, as described above, of the path management table **45** and the volume pair management table **44** held therein to each of the first storage nodes **11** in the primary site **2**, thereby similarly updating the path management table **45** and the volume pair management table **44** held by each of the first storage nodes **11**. Then, the secondary site management software **34** transmits differential data of before and after the update, as described above, of the path management table **45** and the volume pair management table **44** held therein to each of the second storage nodes **30** in the secondary site **3**, thereby similarly updating the path management table **45** and the volume pair management table **44** held by each of the second storage nodes **30**.

[0239] As described above, the remote path **7** is set between the standby specific storage node **11** and the standby movement destination storage node **30**. Note that the primary site management software **26** and the secondary site management software **34** may further cooperate to create an iSCSI initiator associated with the primary volume PVOL corresponding to the selected standby

secondary volume SVOL in the standby specific storage node **11**, create an iSCSI target associated with the selected standby secondary volume SVOL in the standby movement destination storage node **30**, and perform processes until a session between the iSCSI initiator and the iSCSI target is established.

[0240] Next, the pair management software **6** collects, from the primary site management device **12**, information necessary for deleting the remote copy path **7** connecting the decreased storage node **30**, which is the movement source of the selected standby secondary volume SVOL, and the standby specific storage node **11** (S39). Specifically, the pair management software **6** extracts the node ID of the decreased source storage node **30** from the rearrangement information notified from the secondary site management software **34**, and inquires the primary site management software **26** about the IP address of each port **23** included in the decreased storage node **30** to which the node ID is assigned.

[0241] The primary site management software **26** that has received this inquiry acquires the port ID of each port **23** included in the decreased storage node **30** from the record corresponding to the decreased storage node **30** in the system configuration table **40** (FIG. 5) held therein. Then, the primary site management software **26** acquires the IP address of each port **23** of the decreased storage node **30** that has acquired the port ID in the above manner from the port management table **42** (FIG. 7) held therein. The primary site management software **26** then transmits the acquired IP address of each port **23** of the acquired decreased storage node **30** to the pair management software **6**.

[0242] Then, the pair management software **26** gives an instruction to the primary site management software **26** and the secondary site management software **34** to delete the remote copy path **7** connecting the standby specific storage node **11** and the decreased storage node **30** based on the IP address of each port **23** included in the decreased storage node **30** acquired in the above manner and the IP address of each port **23** included in the standby specific storage node **11** acquired in step S32 (S41). Note that this instruction includes information on the IP address of each port **23** included in the standby specific storage node **11** and information on the IP address of each port **23** included in the decreased storage node **30**.

[0243] Thus, the primary site management software **26** and the secondary site management software **34** that have received this instruction specify a record in which at least one of the combinations of the IP address of each port **23** included in the standby specific storage node **11** and the IP address of each port **23** included in the decreased storage node **30** is stored in the first or second path fields **45CA** and **45CB** from the records of the path management table **45** held therein, and delete the record from the path management table **45**.

[0244] In addition, the primary site management software **26** transmits differential data of before and after the update as described above of the path management table **45** held therein to each of the first storage nodes **11** in the primary site **2**, thereby similarly updating the path management table **45** held by each of the first storage nodes **11**. Furthermore, the secondary site management software **34** also transmits differential data of before and after the update as described above of the path management table **45** held therein to each of the second storage nodes **30** in the secondary site **3**, thereby similarly updating the path management table **45** held by each of the second storage nodes **30**. As described above, the remote path **7** set between the standby specific storage node **11** and the decreased storage node **30** is deleted.

[0245] Thereafter, the pair management software **26** determines whether or not the processes after step S38 have been finished execution for all the active secondary volumes SVOL moved from the decreased storage node **30** recognized from the rearrangement information notified from the secondary site management software **34** to the other second storage nodes **30** (S42). Then, when a negative result is obtained in this determination, the pair management software **26** returns to step S37, and thereafter, repeats the processes of steps S37 to S42 while sequentially switching the standby secondary volume SVOL selected in step S37 to another corresponding standby secondary

volume SVOL which is not processed after step S38.

[0246] By this repetition process, the remote copy path 7 connecting each standby secondary volume SVOL moved from the decreased storage node 30 to another second storage node 30 and the primary volume PVOL corresponding to the standby secondary volume SVOL is sequentially set between the corresponding standby specific storage node 11 and the standby movement destination storage node 30, and the original remote copy path 7 set between the decreased storage node 30 that is the movement source of the standby secondary volume SVOL and the standby specific storage node 11 is sequentially deleted.

[0247] Then, the pair management software 6 ends the execution of the processes of step S38 to step S41 for each standby secondary volume SVOL moved from the decreased storage node 30 to another second storage node 30 in due course, and when a positive result is obtained in step S42, the pair management software side storage node decreasing process is ended.

(4) EFFECTS OF PRESENT EMBODIMENT

[0248] As described above, in the storage system 1 of the present embodiment, when the number of the second storage nodes 30 is increased/decreased in the site 3, under the control of the pair management device 5, a new remote copy path 7 is reset to the second storage node 30 of a rearrangement destination of the storage control software 35 rearranged along with the increase/decrease of the second storage node 30 and the secondary volume SVOL for which the storage control software 35 has the ownership, and the corresponding original remote copy path 7 set so far in the second storage node 30 of a rearrangement source is deleted.

[0249] Therefore, in the present storage system 1, even after the increase/decrease of the second storage node 30, remote copy between the second storage node 30 of the rearrangement destination of the rearranged storage control software 35 and the secondary volume SVOL for which by the storage control software 35 has the ownership, and the first storage node 11 in which the primary volume PVOL associated with the secondary volume SVOL is arranged can be performed via the newly created remote copy path 7.

[0250] Therefore, according to the present storage system 1, even in a case where a failure occurs in the active storage control software 35 or the second storage node 30 in which the active storage control software 35 is arranged and the failover is executed, it is possible to suppress the transfer of the journal data JNL in the secondary site 3 due to the increase/decrease of the second storage node 30, and thus, it is possible to prevent the performance degradation of the remote copy process.

(5) OTHER EMBODIMENTS

[0251] Note that, in the above-described embodiment, a case where the primary site management device 12 is provided separately from the first storage node 11 in the primary site 2 has been described, but the present invention is not limited thereto, and the function of the primary site management device 12 may be implemented in any of the first storage nodes 11, and the primary site management device 12 may be omitted.

[0252] Similarly, in the above-described embodiment, a case where the secondary site management device 31 is provided separately from the second storage node 30 in the secondary site 3 has been described, but the present invention is not limited thereto, and the function of the secondary site management device 31 may be implemented in any of the second storage nodes 30, and the secondary site management device 31 may be omitted.

[0253] Furthermore, in the above-described embodiment, a case where the present invention is applied to a case where the second storage node 30 is increased/decreased in the secondary site 3 has been described, but the present invention is not limited thereto, and can be similarly applied to a case where the first storage node 11 is increased/decreased in the primary site 2.

[0254] Furthermore, in the above-described embodiment, the case where the pair management software 6 acquires necessary information from the primary site management device 5 in steps S12 and S14 of the pair management software side storage node increasing process and steps S32 and S34 of the pair management software side storage node decreasing process has been described, but

the present invention is not limited thereto, and the information may be acquired from the secondary site management device **31** or the first or second storage nodes **11** and **30**. In addition, the information on the primary site **2** may be acquired from the primary site management device **5**, and the information on the secondary site **3** may be acquired from the secondary site management device **31**.

[0255] Furthermore, in the above-described embodiment, a case where the pair management device **5** includes one computer device has been described, but the present invention is not limited thereto, and the pair management device **5** may include a distributed computing system including a plurality of computer devices.

[0256] The present invention can be applied to a storage system including one or a plurality of first storage nodes installed at a primary site and a plurality of second storage nodes installed at a secondary site.

Claims

1. A storage system including one or a plurality of first storage nodes installed at a primary site and a plurality of second storage nodes installed at a secondary site, wherein the first storage node is provided with a primary volume provided to a host device as a storage area for reading and writing data, the second storage node is provided with a secondary volume as a storage area for backing up the data written in the corresponding primary volume, a remote copy path for remotely copying the data written in the primary volume to the secondary volume is set between the first storage node and the second storage node provided with the secondary volume corresponding to the primary volume provided in the first storage node, and a first management device that sets, when the second storage node is increased/decreased at the secondary site, a new remote copy path between the first storage node provided with the primary volume associated with the secondary volume moved in accordance with the increase/decrease, and deletes an original remote copy path set between the second storage that is a movement source of the secondary volume moved in accordance with the increase/decrease and the first storage node.
2. The storage system according to claim 1, wherein the first management device is installed at a location different from the primary site and the secondary site.
3. The storage system according to claim 1, further comprising: a second management device that is disposed at the primary site and manages each of the first storage nodes installed at the primary site; and a third management device that is disposed at the secondary site and manages each of the second storage nodes installed at the secondary site, wherein the first management device includes, collecting necessary information from the second and/or third management devices when the second storage node is increased/decreased at the secondary site, and instructing the second and third management devices to set the new remote path and delete an original second remote copy path based on the collected information.
4. The storage system according to claim 3, wherein the first management device, acquires, when the second storage node is increased, information of the secondary volume moved to the increased second storage node, information of the second storage node of a rearrangement source of the secondary volume, and information of the increased second storage node from the third management device, acquires an address of the first storage node in which the primary volume corresponding to the secondary volume moved to the increased second storage node exists and an address of the increased second storage node from the second or third management device based on the acquired information, and instructs the second and third management devices to set a new remote copy path between the first storage node and the increased second storage node based on the acquired addresses.
5. The storage system according to claim 4, wherein the first management device, acquires an address of the second storage node of a movement source of the secondary volume moved to the

increased second storage node from the second or third management device based on the information acquired from the third management device, and instructs the second and third management devices to delete an original remote copy path between the first storage node in which the primary volume corresponding to the moved secondary volume exists and the second storage node of the movement source of the secondary volume based on the acquired address.

6. The storage system according to claim 3, wherein the first management device, acquires, from the third management device, information on the secondary volume moved from the decreased second storage node to another second storage node, information on the second storage node to which the secondary volume is moved, and information on the decreased second storage node when the second storage node is decreased; acquires, from the second or third management device, an address of the first storage node in which the primary volume associated with the secondary volume moved from the decreased second storage node to another second storage node exists and an address of the second storage node to which the secondary volume is moved from the decreased second storage node based on each acquired information; and instructs the second and third management devices to set a new remote copy path between the first storage node in which the primary volume corresponding to the moved secondary volume exists and the second storage node of a movement destination of the secondary volume based on the acquired addresses.

7. The storage system according to claim 6, wherein the first management device acquires an address of the decreased second storage node from the second or third management device based on the information acquired from the third management device, and instructs the second and third management devices to delete an original remote copy path between the first storage node in which the primary volume corresponding to the moved secondary volume exists and the decreased second storage node based on the acquired address.

8. A path management method executed in a storage system including one or a plurality of first storage nodes installed at a primary site and a plurality of second storage nodes installed at a secondary site, wherein the first storage node is provided with a primary volume provided to a host device as a storage area for reading and writing data, the second storage node is provided with a secondary volume as a storage area for backing up the data written in the corresponding primary volume, a remote copy path for remotely copying the data written in the primary volume to the secondary volume is set between the first storage node and the second storage node provided with the secondary volume corresponding to the primary volume provided in the first storage node, the storage system includes a first management device, a second management device that is disposed at the primary site and manages each of the first storage nodes installed at the primary site, and a third management device that is disposed at the secondary site and manages each of the second storage nodes installed at the secondary site, the method including, a first step in which the first management device collects necessary information from the second and/or third management device when the second storage node is increase/decreased at the secondary site; and a second step in which the first management device instructs the second and third management devices to set a new remote copy path between the first storage node provided with the primary volume associated with the secondary volume moved in accordance with the increase/decrease based on the collected information, and delete an original remote copy path set between the second storage of a movement source of the secondary volume moved in accordance with the increase/decrease and the first storage node.

9. The path management method according to claim 8, wherein in the first step, the first management device acquires, when the second storage node is increased, information of the secondary volume moved to the increased second storage node, information of the second storage node of a rearrangement source of the secondary volume, and information of the increased second storage node from the third management device, acquires an address of the first storage node in which the primary volume corresponding to the secondary volume moved to the increased second storage node exists and an address of the increased second storage node from the second or third

management device based on the acquired information, and in the second step, the first management device instructs the second and third management devices to set a new remote copy path between the first storage node and the increased second storage node based on the acquired addresses.

10. The path management method according to claim 9, wherein in the first step, the first management device acquires an address of the second storage node of a movement source of the secondary volume moved to the increased second storage node from the second or third management device based on the information acquired from the third management device, and in the second step, the first management device instructs the second and third management devices to delete an original remote copy path between the first storage node in which the primary volume corresponding to the moved secondary volume exists and the second storage node of a movement source of the secondary volume based on the acquired address.

11. The path management method according to claim 8, wherein in the first step, the first management device acquires, from the third management device, information on the secondary volume moved from the decreased second storage node to another second storage node, information on the second storage node to which the secondary volume is moved, and information on the decreased second storage node when the second storage node is decreased, and acquires, from the second or third management device, an address of the first storage node in which the primary volume associated with the secondary volume moved from the decreased second storage node to another second storage node exists and an address of the second storage node to which the secondary volume is moved from the decreased second storage node based on each acquired information; and in the second step, the first management device instructs the second and third management devices to set a new remote copy path between the first storage node in which the primary volume corresponding to the moved secondary volume exists and the second storage node of a movement destination of the secondary volume based on the acquired addresses.

12. The path management method according to claim 11, wherein in the first step, the first management device acquires an address of the decreased second storage node from the second or third management device based on the information acquired from the third management device, and in the second step, the first management device instructs the second and third management devices to delete an original remote copy path between the first storage node in which the primary volume corresponding to the moved secondary volume exists and the decreased second storage node based on the acquired address.
